

Q.20 Describe the impact of tolerance of manufacturing.
(CO4)

Q.21 What are the benefits of proper process planning?
(CO4)

Q.22 List and explain steps in cost estimation. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain in detail the various elements involved in engineering design with suitable examples. (CO2)

Q.24 Discuss the overall effects of material properties and manufacturing processes on design decisions.(CO3)

Q.25 Describe the steps involved in process planning and cost estimation in detail. (CO5)

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6th Sem. / Automation & Robotics

Subject : Product Design and Process Planning

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is NOT a general design consideration? (CO1)

- a) Strength b) Appearance
- c) Entertainment d) Functionality

Q.2 A good product design should (CO2)

- a) Be expensive b) Satisfy user needs
- c) Use rare materials
- d) Require complex processes

Q.3 Yield strength of a material affects its (CO2)

- a) Appearance
- b) Electrical conductivity
- c) Elastic limit d) Ductility

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- Q.4 The process planning function is mainly concerned with (CO4)
- a) Cost estimation b) Material selection
c) Product launching d) Process sequence
- Q.5 Which of the following is a primary cost in cost estimation? (CO5)
- a) Labor cost b) Advertising cost
c) Overhead cost d) Packaging cost
- Q.6 Ductility is the ability of a material to (CO2)
- a) Resist corrosion b) Conduct electricity
c) Be drawn into wires d) Absorb shock

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is meant by engineering design? (CO1)
- Q.8 Name two important thermal properties of materials. (CO2)
- Q.9 Define product development. (CO1)
- Q.10 List any one example of ferrous material. (CO2)

- Q.11 What is the purpose of cost estimation? (CO5)
- Q.12 What is the role of material properties in design? (CO3)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the stages involved in product development. (CO1)
- Q.14 Describe general consideration to be taken while designing a product. (CO2)
- Q.15 List and explain any four important mechanical properties of materials (CO2)
- Q.16 How do manufacturing processes affect product design? (CO3)
- Q.17 Explain the role of material selection in engineering design. (CO2)
- Q.18 Write short notes on the phases of engineering design. (CO3)
- Q.19 Explain the relationship between design and cost. (CO5)